

Figure 5: The performance of R-Tuning-R on ParaRel and MMLU datasets. ID and OOD denote in-domain and out-of-domain test datasets, respectively.

Dataset	Model	R-Tuning-R	R-Tuning	Vanilla	Pretrain-T
FalseQA	OpenLLaMA-3B	**98.31**	87.32	2.07	9.98
	LLaMA-7B	**97.67**	96.62	18.35	8.92
	LLaMA-13B	**99.07**	95.90	6.00	24.10
NEC	OpenLLaMA-3B	**99.90**	95.72	0.96	7.31
	LLaMA-7B	**99.52**	99.18	20.55	2.02
	LLaMA-13B	**99.90**	98.17	2.36	4.76
SA	OpenLLaMA-3B	**99.22**	90.99	5.23	18.90
	LLaMA-7B	**98.55**	95.45	34.79	16.96
	LLaMA-13B	**99.71**	96.61	12.21	28.00

Table 7: The refusal rate (%) of R-Tuning and R-Tuning-R, and other baselines on the refusal benchmarks. SA is the unanswerable part of the SelfAware dataset. The refusal rate of R-Tuning-R on the unanswerable datasets is extremely high, while the refusal rate of other fine-tuned methods and pre-trained models is low.

padding, we can directly replace the label words with uncertainty expressions for uncertain questions and keep the original label words for certain questions, which is called the replacement strategy, leading to a variant R-Tuning-R. For example, the certain part of the training questions D_1 is constructed as follows:

$$Q : \{\text{Question}\}, A : \{\text{Answer}\}, \tag{9}$$

while the uncertain dataset D_0 is constructed as follows:

$$Q : \{\text{Question}\}, A : \{\text{Uncertainty Expression}\}. \tag{10}$$

There are many different ways for the uncertainty expression. To increase the diversity, we take the 16 expressions of uncertainty text from Yin et al. (2023). These 16 expressions are listed in the Appendix Section A.5.

We conduct experiments with R-Tuning-R on ParaRel and MMLU datasets by comparing it with vanilla fine-tuning strategy and the original pre-trained models. The results are shown in Figure 5. Firstly, on both in-domain and out-of-domain test sets, the accuracy of R-Tuning-R is higher than Pretrain-T, which benefits from only answering certain questions. More detailed results with answer rate are reported in Table 8, where we find the model is able to refuse a certain amount of questions. Then, R-Tuning-R outperforms Vanilla with a significantly higher accuracy on its willingly answered questions, which demonstrates the effectiveness of our method. It is promising as R-Tuning-R is trained with fewer ground-truth labels, while Vanilla is trained on all labels of the full training data. Generally, larger models possess more powerful refusal abilities. In Figure 5, we observe that on the willingly answered questions, larger models achieve a higher accuracy. In addition, the high accuracy of Pretrain-W reveals that those selected questions are within parametric knowledge of the pre-trained model. In summary, compared with vanilla fine-tuning, R-Tuning-R provides the model with the refusal ability to refuse unknown questions, which eventually improves the accuracy and prevents them from making hallucinated answers. Table 9 shows the case studies of how R-Tuning-R works. There are significant differences when they encounter questions out of their knowledge. The Vanilla model is proactive in making up an answer, which is a hallucination and makes no sense. However, R-Tuning-R refuses them explicitly with keywords *do not know, not known, and impossible*. The ability of R-Tuning-R to refuse unknown questions results in fewer hallucinations.

Despite this refusal ability, there are two issues with R-Tuning-R: (1) the replacement method throws away valuable labels which could be leveraged for training. (2) R-Tuning could either only output the answer or only output the certainty, but cannot respond to both, leading to difficulties in considering the precision and recall simultaneously. To leverage all ground-truth labels during the tuning process, and instruct models to predict answers and express uncertainty at the same time, we employ the padding strategy in our main approach,

Dataset	Domain	Models	R-Tuning-R (%)	Answer Rate (%)	Vanilla (%)	Pretrain-T (%)	Pretrain-W (%)
ParaRel	In-Domain	OpenLLaMA-3B	**82.79**	44.65	60.58	36.23	71.48
		LLaMA-7B	**85.95**	44.11	63.72	37.79	72.23
		LLaMA-13B	**87.06**	44.00	66.53	41.53	78.51
	Out-of-Domain	OpenLLaMA-3B	44.04	40.80	38.68	30.18	**46.73**
		LLaMA-7B	**69.54**	28.07	43.38	34.44	63.09
		LLaMA-13B	**76.13**	30.16	41.82	34.48	68.99
MMLU	In-Domain	OpenLLaMA-3B	**21.99**	5.79	21.12	21.22	18.44
		LLaMA-7B	**46.35**	9.56	33.25	26.77	41.20
		LLaMA-13B	**57.47**	42.52	42.97	41.41	55.54
	Out-of-Domain	OpenLLaMA-3B	24.55	2.41	23.93	25.44	**27.27**
		LLaMA-7B	**55.56**	12.96	38.56	31.09	44.01
		LLaMA-13B	**67.31**	48.32	51.19	47.60	62.53

Table 8: Detailed performance of R-Tuning-R on ParaRel and MMLU dataset. The answer rate means the percentage of willingly answered questions of R-Tuning-R.

Input Questions	**R-Tuning-R**	**Vanilla**	**Ground-Truth**
What field does Lee Alvin DuBridge work in?	I **do not know** the answer.	Music.	Physics.
Where was Blaine Willenborg born?	It is **not known**.	New York.	Miami
Where did Hippolyte Le Bas die?	It is **impossible** to know.	Lyon	Paris

(a) Examples of R-Tuning-R refusing questions that are out of its parametric knowledge. R-Tuning-R expresses its unknown when it does not know the answer. Vanilla produces hallucinated answers when it does not know the answer.

Input Questions	**R-Tuning-R**	**Vanilla**	**Ground-truth**
Where is Lion Air headquartered?	Jakarta.	Jakarta.	Jakarta.
What does Jacobo Zabludovsky work as?	journalist.	journalist.	journalist.
What is the native language of Joseph Conombo?	French.	French.	French.

(b) Examples of R-Tuning-R answering questions within parametric knowledge.

Table 9: Case study of refused and willingly answered questions with R-Tuning-R and Vanilla.

where every question is appended with the ground-truth label and the uncertainty expression, indicating whether the model is confident or not.

A.4 Case Studies of R-Tuning-R

In this section, we display the detailed statistics in Table 8, and illustrate more case studies of R-Tuning-R in Table 9.

A.5 Uncertainty Text

In this section, we list the 16 uncertainty expressions from Yin et al. (2023):

1. The answer is unknown.
2. The answer is uncertain.
3. The answer is unclear.
4. There is no scientific evidence.
5. There is no definitive answer.
6. There is no right answer.
7. There is much debate.
8. There is no known case.
9. There is no concrete answer to this question.
10. There is no public information available.
11. It is impossible to know.
12. It is impossible to answer.
13. It is difficult to predict.
14. It is not known.
15. We do not know.
16. I'm not sure.

Dataset	Model	D_1	D_0
ParaRel	OpenLLaMA-3B	57.92	63.08
	LLaMA-7B	45.81	52.08
	LLaMA-13B	42.79	48.75
MMLU	OpenLLaMA-3B	32.95	462.36
	LLaMA-7B	22.20	115.87
	LLaMA-13B	22.12	81.41
WiCE	OpenLLaMA-3B	61.28	203.43
	LLaMA-7B	20.93	19.40
	LLaMA-13B	17.73	19.56
HotpotQA	OpenLLaMA-3B	144.89	170.38
	LLaMA-7B	49.97	60.19
	LLaMA-13B	42.60	55.20
FEVER	OpenLLaMA-3B	88.38	72.11
	LLaMA-7B	38.46	43.69
	LLaMA-13B	39.00	44.14

Table 10: Perplexity of the training datasets. We run the pre-trained models on the context and questions and calculate the average perplexity.

Dataset	Model	D_1	D_0
ParaRel	OpenLLaMA-3B	0.426	0.709
	LLaMA-7B	0.475	0.694
	LLaMA-13B	0.436	0.744
MMLU	OpenLLaMA-3B	0.347	0.389
	LLaMA-7B	0.330	0.400
	LLaMA-13B	0.239	0.457
WiCE	OpenLLaMA-3B	0.250	0.280
	LLaMA-7B	0.254	0.270
	LLaMA-13B	0.265	0.252
HotpotQA	OpenLLaMA-3B	0.534	0.747
	LLaMA-7B	0.605	0.719
	LLaMA-13B	0.528	0.797
FEVER	OpenLLaMA-3B	0.413	0.219
	LLaMA-7B	0.279	0.286
	LLaMA-13B	0.189	0.350

Table 11: Entropy of the training datasets. It is calculated from the frequency of every predicted answer among all predictions. A larger entropy denotes greater uncertainty of the system.

A.6 Perplexity of Datasets

Perplexity measures how well the language model predicts a given text. Lower perplexity means better prediction and understanding of the text. According to the refusal-aware data identification, we split the training data into two sets: D_0 (uncertain questions) and D_1 (certain questions). To uncover why the pre-trained model responds to them differently, we calculate the average perplexity on these two datasets with the pre-trained models. The perplexity is calculated as follows:

$$\text{PPL}(X) = \exp\left\{-\frac{1}{t}\sum_{i=1}^{t}\log p_\theta(x_i \mid x_{<i})\right\}, \tag{11}$$

where X denotes a sentence consisting of tokens and $X = (x_1, x_2, \ldots, x_t)$. Specifically, we calculate the perplexity of the training questions to estimate the pre-trained model's understanding of them. The results are shown in Table 10. We observe that D_1 has a lower perplexity, demonstrating that the pre-trained model is more familiar with the questions and is likely to provide the correct answer. For D_0, its higher perplexity shows that these questions are not familiar to the model and out of the model's knowledge, and this is the reason why the model tends to hallucinate text instead of providing the correct answers. We also observe that larger models have a lower perplexity and randomness on the questions, which is why larger models generally perform better on various tasks.

By instructing our model to express uncertainty toward relatively random questions in terms of perplexity, the model develops a better understanding of uncertainty and ambiguity and learns the ability to recognize when it does not know. This ability is crucial in situations where simply providing a definite answer may be inappropriate or even harmful. On the other hand, since our model is also trained with data with certain expressions, it becomes more proficient at handling less random questions, and answering them with confidence and certainty. Overall, R-Tuning improves the model's ability to adapt to different levels of question randomness.

To verify the pre-trained model is less familiar with the uncertain questions while more confident with certain questions, we also plot the confidence distribution on certain questions and uncertain questions, shown in Figure 7 in Appendix A.9. It is observed that a larger percentage of certain questions occupies the high confidence intervals, which means when the model provides correct answers, it generally shows larger confidence.